CTF Challenge Write-up: " X fOR xylographs"

Category: Forensics

Walnut drew a selfie but got super shy about showing it. So instead of sharing it, he locked it up with an 8-character key. Unfortunately, Walnut forgot his key! Can you help him find it out and decrypt the image?

Challenge Summary

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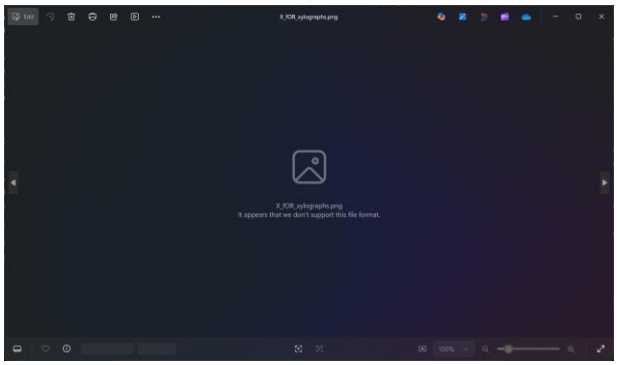
In this challenge, we were given a corrupted image file that would not open normally. By inspecting the file in a hex editor, I noticed the header bytes did not match the expected PNG signature. The challenge title hinted at XOR encryption, so I found the correct XOR key (` 73 6B 65 74 63 68 65 73`) and use CyberChef to restore the file to a valid PNG. After analysising with ExifTool, I found that the file was actually an APNG. Extracting its frames exposed a hidden second layer, which contained the correct flag.

Detailed Walkthrough

Step 1: Initial Investigation

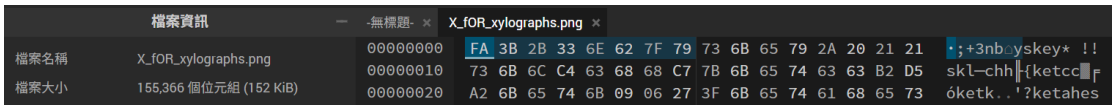
After downloading the provided image (X_fOR_xylographs.png), I noticed it could not be opened with a normal image viewer. This suggested that the file might be corrupted or intentionally obfuscated.



Step 2: Hex Analysis

I opened the file in Hexed.it (https://hexed.it/) to inspect the raw bytes. At offset 0x00000000, the file began with:

FA 3B 2B 33 6E 62 7F 79



By checking CTF Wiki (https://ctf-wiki.org/zh-tw/misc/picture/png/?h=png)

We found that a valid PNG file should start with the standard header:

89 50 4E 47 0D 0A 1A 0A

So if we change FA 3B 2B 33 6E 62 7F 79 to 89 50 4E 47 0D 0A 1A 0A, then we may open the file with a normal image viewer. This means we can further analyze the data

Step 3: Considering XOR Obfuscation

But how should we do this?

The challenge title, " X fOR xylographs ", hinted at the use of XOR encryption. To test this, I compared the given bytes with the expected PNG header.

We known that XOR can be reversible. If you know any two of the three values (Original, Corrupted, Key), you can always find the third:

- Original = Corrupted $\oplus$ Key
- Key = Corrupted $\oplus$ Original

FA	$\oplus$	89	= 73
3B	$\oplus$	50	= 6B
2B	$\oplus$	4E	= 65
33	$\oplus$	47	= 74
6E	$\oplus$	0D	= 63
62	$\oplus$	0A	= 68
7F	$\oplus$	1A	= 65
79	$\oplus$	0A	= 73

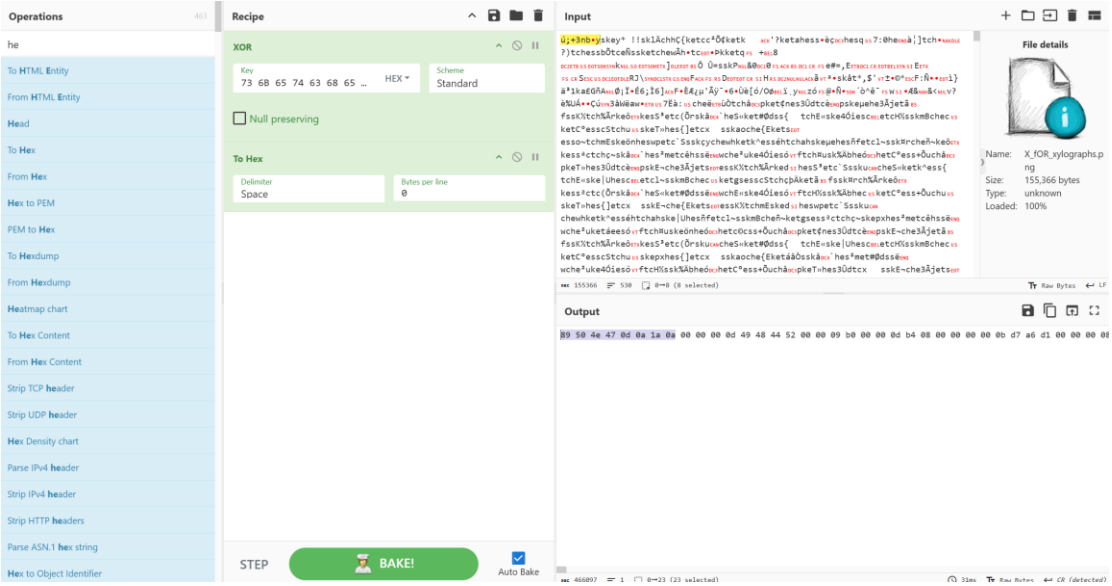
now we have Corrupted and Original

Corrupted = FA 3B 2B 33 6E 62 7F 79

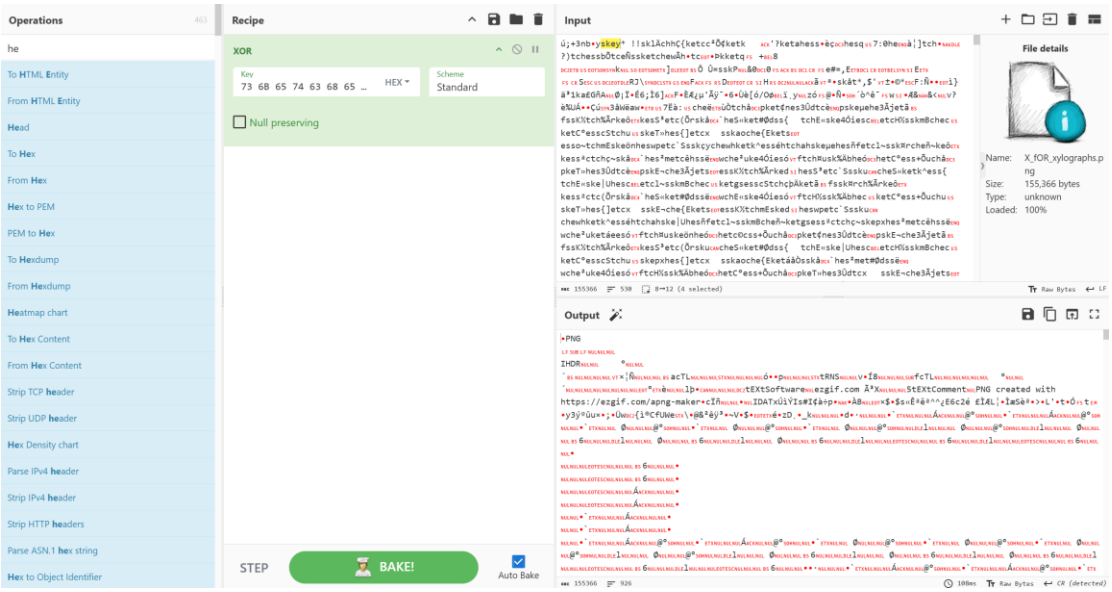
Original = 89 50 4E 47 0D 0A 1A 0A

So the key is: “73 6B 65 74 63 68 65 73”

Using CyberChef XOR operation and applying this key, I recovered the correct PNG signature. (use To Hex to have a check)



Remember to remove "To hex" or your image will still not open after download.



Now you can save the file and output as a png file

In fact, you can use artificial intelligence to help you find the key, so you can spend less time on XOR (just ask AI how to change FA 3B 2B 33 6E 62 7F 79 to 89 50 4E 47 0D 0A 1A 0A by using XOR)

Step 5: Opening the File

With the corrected header, the file could now be opened as a PNG. However, no obvious flag or hidden message was visible in the image itself.

Maybe I should check its exif data to get more information.

Step 6: Metadata Analysis

ExifTool output:

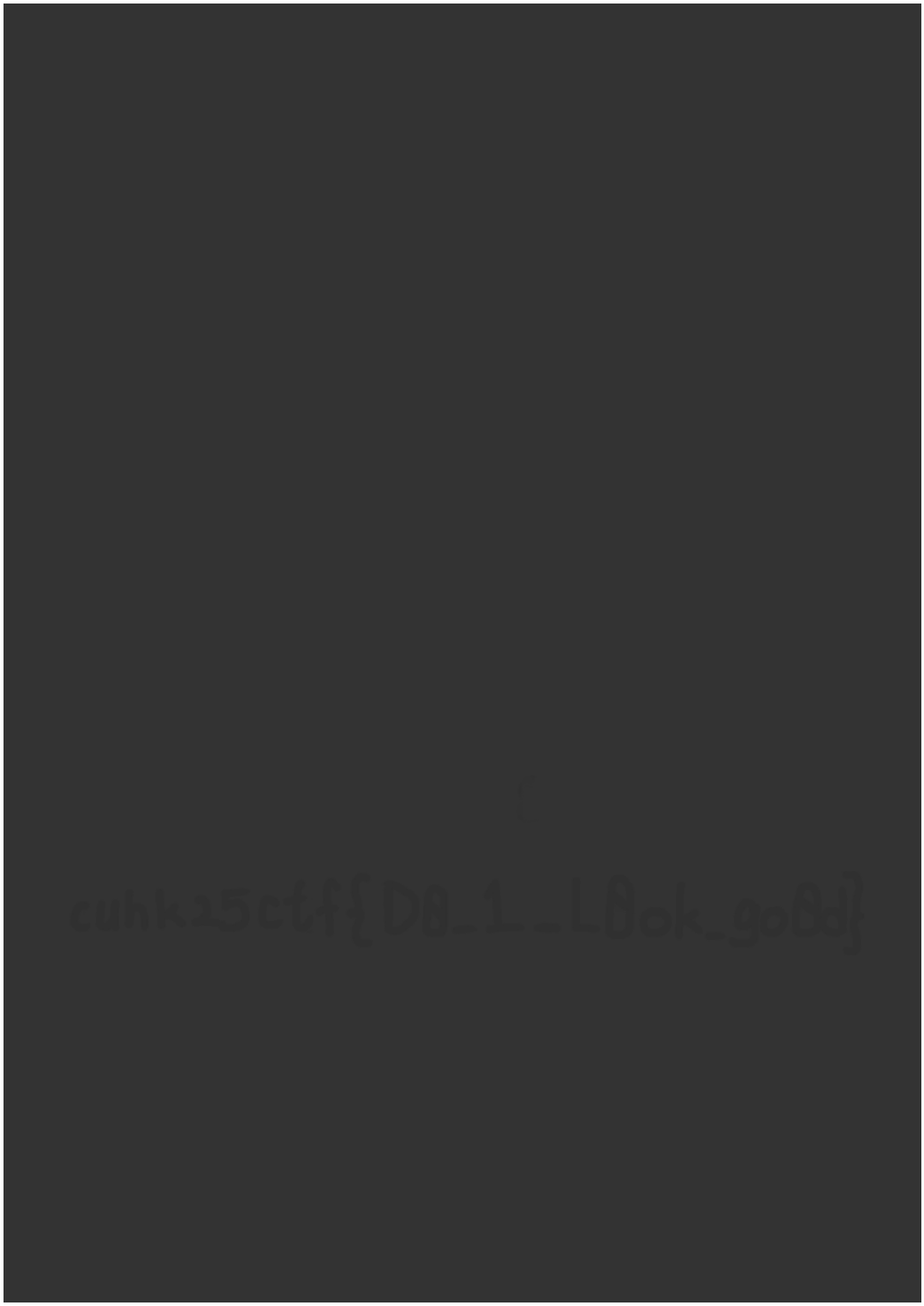
ExifTool Version Number	12.72
File Name	file.png
Directory	.
File Size	155 kB
File Modification Date/Time	2025:10:05 07:35:51+02:00
File Access Date/Time	2025:10:05 07:35:51+02:00
File Inode Change Date/Time	2025:10:05 07:35:51+02:00
File Permissions	-rw-r--r--
File Type	APNG
File Type Extension	png
MIME Type	image/apng
Image Width	2480
Image Height	3508
Bit Depth	8
Color Type	Grayscale
Compression	Deflate/Inflate
Filter	Adaptive
Interlace	Noninterlaced
Animation Frames	2
Animation Plays	inf
Transparency	0
Software	ezgif.com
Comment	PNG created with https://ezgif.com/apng-maker
Warning	[minor] Trailer data after PNG IEND chunk
Image Size	2480x3508
Megapixels	8.7

I uploaded the file to ezgif (https://ezgif.com/view-metadata), an online ExifTool service, to inspect its metadata. This revealed that the file was actually an **APNG** (Animated PNG), meaning it contained multiple frames.

The flag may be stored in another frame.

Step 7: Extracting Frames

Since APNGs can store multiple layers, I used an online GIF/APNG frame splitter (https://ezgif.com/split) to separate the frames. The second frame contained the hidden data.



Can u see? LOL

Step 8: Retrieving the Flag

Launch **Painter** and use the **Paint Bucket Tool** to turn the black blackboard white — then you'll see what appears.

Now you have found the right flag ヽ(≧▽≦)o*